A third-moment obstruction to the claimed free CLT rate

Disproof of the universal moment-rate clause in TLMC 00000007730

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Exact scope. The source claims an $O(n^{-1})$ moment-convergence rate for centered self-adjoint tuples free over B with bounded fourth moments. Neither language assumes symmetry or a vanishing third moment. We use the scalar specialization $B = \mathbb{C}$ and one-entry tuples. Only this universal rate clause is refuted. The complete bilingual statement is preserved in `SOURCE.md`. We make no assertion about a theorem restricted to even moments, or the later fourth-cumulant and colored-Motzkin clauses.

Theorem 1. *There is a concrete unital star-algebra $\mathcal{A}$ of bounded operators on a complex Hilbert space, with a positive unital bounded tracial state τ, and a freely independent identically distributed sequence of bounded self-adjoint elements $Y_i \in \mathcal{A}$ such that*

$$\tau(Y_i) = 0, \quad \tau(Y_i^2) = 1, \quad \tau(Y_i^4) = \frac{3}{2}, \qquad S_n = \frac{1}{\sqrt{n}} \sum_{i=0}^{n-1} Y_i.$$

For every positive integer n,

$$\tau(S_n^3) = \frac{1}{\sqrt{2n}}, \qquad \int x^3 \, d\mu_{\mathrm{sc}}(x) = 0,$$

where μ_{sc} is the actual standard semicircle probability measure. The discrepancy is not $O(n^{-1})$ as $n \to \infty$.

1. The actual free product, Hilbert space and trace. Let $F_i = C_3 = \langle g_i \mid g_i^3 = 1 \rangle$ and $G = *_{i \in \mathbb{N}} F_i$. Lean uses the native countable indexed group coproduct and its genuine factor embeddings. Its reduced-word equivalence proves the normal-form theorem; countability follows from the actual list of factor letters. Set $H = \ell^2(G; \mathbb{C})$, using native `lp` with exponent 2. Actual coordinate reindexing gives linear isometry equivalences

$$(\lambda(g)f)(h) = f(g^{-1}h).$$

Their bounded operators satisfy $\lambda(g)\lambda(h) = \lambda(gh)$ and $\lambda(g)^* = \lambda(g^{-1})$, with both unitary identities proved.

The universal group-algebra map is the actual algebra homomorphism $\rho : \mathbb{C}[G] \to \mathcal{B}(H)$ obtained from these operators. For $\delta = \delta_e$, Lean proves the coefficient identity

$$(\rho(a)\delta)(h) = a_h.$$

Thus ρ is faithful. Its image is closed under the genuine adjoint and forms a native star-subalgebra $\mathcal{A}$ of $\mathcal{B}(H)$. This is the concrete unital star-algebra of bounded operators used here, in the standard algebraic formulation of noncommutative probability [1, p. 332].

On this actual algebra define $\tau(T) = \langle \delta, T\delta \rangle$. The formal proof establishes

$$\tau(1) = 1, \quad |\tau(T)| \le \|T\|, \quad \tau(T^*T) = \|T\delta\|^2, \quad \tau(AB) = \tau(BA).$$

For traciality, $\tau(\rho(a)) = a_e$ and the equality $(ab)_e = (ba)_e$ is proved for all finitely supported group-algebra elements by their true linear induction principle. All tracial claims concern $\mathcal{A}$. The state is bundled as a genuine continuous linear map.

2. Full freeness. Let $\mathcal{A}_i$ be the actual image of $\mathbb{C}[C_3]$ under the ith factor representation, regarded as a native unital star-subalgebra of $\mathcal{A}$. If $a \in \mathbb{C}[C_3]$ is centered, its identity coefficient is zero. Every element in the support of its embedded image therefore comes from a nonidentity letter of that factor. For an arbitrary product of such elements, the native support-of-product inclusion extracts an actual list of nonidentity factor letters. When consecutive factor indices differ, this list is a nonempty reduced word, so its product cannot equal e by the native normal-form equivalence. Consequently

$$\tau(a_1 \cdots a_k) = 0 \quad (k > 0,\ a_j \in \mathcal{A}_{i_j},\ \tau(a_j) = 0,\ i_j \neq i_{j+1}).$$

This is proved for every length and every centered element of each whole factor algebra. It is then transported to the actual represented star-algebra. The group-algebra coefficient-state construction and this standard freeness criterion are described in [1, pp. 339–340]; the particular C_3 calculations below are proved directly.

3. Genuine variables, identical laws and fourth moments. Put $U_i = \lambda(g_i)$ and $X_i = U_i + U_i^*$. These are the images of the actual factor-group-algebra element $g + g^{-1}$. Their self-adjointness and membership in $\mathcal{A}_i$ are proved, and $U_i^3 = 1$ gives

$$X_i^2 = X_i + 2, \qquad X_i^3 = 3X_i + 2, \qquad X_i^4 = 5X_i + 6.$$

Since neither g nor g^{-1} is the identity of C_3, the true coefficient state gives $\tau(X_i) = 0$. Applying the state to these identities yields the actual values

$$\tau(X_i^2) = 2, \qquad \tau(X_i^3) = 2, \qquad \tau(X_i^4) = 6.$$

No moment is assigned as a definition. The identities are obtained in the native factor group algebra, mapped through the genuine bounded representation, and evaluated by the actual state.

Define $Y_i = X_i/\sqrt{2}$. Real scalar multiplication preserves the proved self-adjointness and factor membership. Direct normalization gives $\tau(Y_i) = 0$, $\tau(Y_i^2) = 1$, and $\tau(Y_i^4) = 3/2$, with the explicit uniform bound $\|\tau(Y_i^4)\| \le 2$. The full polynomial law is also checked: for every $p \in \mathbb{C}[z]$ and all i, j,

$$\tau\big(p(Y_i)\big) = \tau\big(p(Y_j)\big).$$

Indeed evaluation commutes with the actual factor-algebra homomorphism, and the trace of any factor image is its original identity coefficient. Since each variable is self-adjoint, this proves equality of their one-variable star-polynomial distributions. Together with the full factor-algebra freeness above, the sequence is free i.i.d.

4. The actual normalized sum and its third moment. The Lean development forms the genuine finite sums of these bounded operators, and also the same sums in the native star-subalgebra $\mathcal{A}$. It proves that their operator images agree and that

$$S_n = \frac{1}{\sqrt{n}} \sum_{i<n} Y_i = \frac{1}{\sqrt{2n}} \sum_{i<n} X_i.$$

All square roots are native real square roots, embedded in $\mathbb{C}$. For actual indices i, j, k, the exact mixed third moment is

$$\tau(X_i X_j X_k) = \begin{cases} 2, & i = j = k, \\ 0, & \text{otherwise.} \end{cases}$$

If adjacent indices differ, full freeness applies. If exactly two adjacent indices agree, replace their product by $X_i + 2$ and use centering and the two-factor freeness identity. The all-equal case uses the genuine cubic identity. Expanding the true cube of the finite sum and evaluating all three finite sums therefore gives

$$\tau\left(\left(\sum_{i<n} X_i\right)^3\right) = 2n.$$

For $n > 0$, the proved square-root identity $(\sqrt{2n})^2 = 2n$ and its nonzero value imply

$$\tau(S_n^3) = \frac{2n}{(\sqrt{2n})^3} = \frac{1}{\sqrt{2n}}.$$

The final rate statement concerns this actual state of the actual operator sum; a separate sequence with a stipulated value is never substituted.

5. The actual semicircle target. Lean defines the real measure

$$d\mu_{\mathrm{sc}}(x) = \mathbf{1}_{[-2,2]}(x) \frac{\sqrt{4 - x^2}}{2\pi}\, dx$$

by native restriction of Lebesgue measure and `withDensity`. This is the standard variance-one semicircle density [1, p. 327], with the normalization matching the Y_i above. Its normalization is proved, rather than assumed: the change of variables $x = 2t$ reduces its integral to the native theorem

$$\int_{-1}^{1} \sqrt{1 - t^2}\, dt = \frac{\pi}{2}.$$

The resulting instance is a true `IsProbabilityMeasure`. The density is continuous and nonnegative. Continuity on the actual compact support also proves the cubic integrand is genuinely integrable against this measure. Finally, its weighted cubic integrand is odd; native interval reflection and linearity give $\int x^3\, d\mu_{\mathrm{sc}} = 0$. This is not a default value of an undefined or nonintegrable Bochner integral.

6. Quantified failure of the claimed rate. Let

$$E_n = \tau(S_n^3) - \int x^3 \, d\mu_{\mathrm{sc}}(x) \in \mathbb{C}.$$

The two actual calculations above prove $\|E_n\| = 1/\sqrt{2n}$ for every $n > 0$. Given any $C \geq 0$ and starting index N, choose a natural number $n \geq N$ with $n > 0$ and $n > 2C^2$. Then

$$(C\sqrt{2n})^2 = 2C^2 n < n^2, \qquad \frac{C}{n} < \frac{1}{\sqrt{2n}} = \|E_n\|.$$

All factors in the squaring and division steps are proved nonnegative or positive. Lean proves the full quantifier statement

$$\forall C \geq 0 \; \forall N \in \mathbb{N} \; \exists n \geq N : \quad n > 0 \quad \text{and} \quad C/n < \|E_n\|.$$

The native eventual-bound characterization then yields

$$\neg \, \mathrm{Asymptotics.IsBigO}_{n \to \infty}\left(E_n, n^{-1}\right).$$

Thus the first universal rate clause in the printed conjecture fails already for scalar-valued, centered, bounded, self-adjoint, free i.i.d. variables with unit variance and uniformly bounded fourth moments. The usual free central limit theorem remains compatible with this example: the third-moment discrepancy does tend to zero. The $O(n^{-1})$ bound asserted without an odd-moment hypothesis is the specific obstruction established here.

Formal objects and reproduction. The complete final theorem is `TLMC7730.complete_counterexample`. The native countable group coproduct, unique reduced words, ℓ^2 Hilbert space, actual bounded regular representation, native image star-subalgebra, bounded positive unital tracial state, whole factor subalgebras and all-length freeness, polynomial laws, real finite sums, actual semicircle probability measure and native asymptotic predicate are connected in the same development. The general free CLT is not needed as a premise for the finite-sum moment calculations or the rate contradiction.

The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

From `lean/`, run the following four commands separately:

```
lean --version
lake build +Main
lake env lean -DwarningAsError=true Check.lean
lake env leanchecker --fresh Main
```

`lean/verify.sh` runs the last three; the version check is separate. The audit calls ordinary `#print axioms` on every owned declaration in the actual imported environment, including generated names. Only `propext`, `Classical.choice` and `Quot.sound` are permitted. The submission retains raw source-bound command logs, execution receipts, actual owned-thread masks and a SHA-256 manifest. Production durations are diagnostic and supply no controlled efficiency conclusion. Organizer review remains pending.

References

[1] J. Novak, with illustrations by M. LaCroix, *Three lectures on free probability*, in *Random Matrices*, MSRI Publications 65 (2014), 309–383. Relevant definitions and models: pp. 327, 329–330, 332, 339–340. Publisher-hosted full text.